

UMBA, Russian Federation

WMO: 223240

Lat: 66.681N

Lon: 34.350E

Elev: 40

StdP: 100.85

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.3	-24.3	-29.6	0.2	-27.3	-26.2	0.3	-24.3	10.1	-2.9	9.0	-6.3	2.2	0	0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	22.8	17.1	21.0	16.1	19.3	15.2	18.2	21.7	17.0	20.0	15.8	18.5	3.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	11.9	19.9	15.5	11.1	18.8	14.4	10.3	17.6	51.5	21.7	47.8	20.0	44.4	18.5	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.6	6.9	-30.4	25.5	4.2	1.9	-33.4	26.9	-35.9	27.9	-38.3	29.0	-41.3	30.3		
(5)			-30.5	19.3	4.2	1.3	-33.5	20.3	-36.0	21.0	-38.3	21.8	-41.4	22.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-10.0	-10.5	-5.8	-0.3	5.1	10.9	14.9	13.0	8.4	2.5	-3.5	-6.8		
	DBStd	9.83	6.88	7.12	4.82	3.88	4.14	3.63	3.35	3.34	3.15	3.62	4.93	6.27		
	HDD10.0	3422	619	574	490	308	161	32	3	8	67	233	405	521		
	HDD18.3	6133	878	807	748	558	410	226	114	170	297	491	655	779		
	CDD10.0	343	0	0	0	0	9	58	155	101	21	0	0	0		
	CDD18.3	13	0	0	0	0	0	1	8	4	0	0	0	0		
	CDH23.3	39	0	0	0	0	0	5	23	11	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	1	0	0	0	0		
(14) Wind	WSAvg	3.4	3.5	3.4	3.6	3.5	3.5	3.4	3.1	3.1	3.3	3.6	3.6	3.6		
(15) Precipitation	PrecAvg	489	27	21	24	25	35	48	55	64	55	54	48	33		
	PrecMax	685	50	55	49	63	91	135	122	181	126	122	121	61		
	PrecMin	329	7	6	3	1	2	3	4	16	3	5	8	0		
	PrecStd	77	11	11	12	13	18	25	32	34	29	30	21	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	3.0	5.0	11.2	19.4	23.7	25.7	24.8	17.6	10.7	6.2	3.6		
		MCWB	1.0	1.9	2.0	6.1	12.3	16.4	18.6	18.8	14.0	9.4	5.5	2.4		
	2%	DB	0.5	0.9	2.8	8.0	15.7	20.4	23.6	21.8	15.6	9.5	4.7	2.2		
		MCWB	0.2	0.3	0.9	4.2	10.1	14.7	17.6	17.0	12.6	8.4	4.1	1.5		
	5%	DB	-0.5	-0.1	1.5	6.1	13.6	18.4	21.8	19.7	14.2	8.4	3.3	1.2		
		MCWB	-0.9	-0.7	0.1	3.1	8.9	13.0	16.6	16.2	12.0	7.5	2.7	0.7		
	10%	DB	-1.6	-1.4	0.4	4.7	11.5	16.7	20.3	17.8	13.0	7.4	2.3	0.3		
		MCWB	-2.0	-1.9	-0.7	2.1	7.7	12.0	15.9	14.7	11.3	6.6	1.7	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	2.0	2.5	6.4	13.0	17.6	19.9	19.6	14.5	10.0	5.7	2.7		
		MCDB	1.5	2.9	4.2	10.0	17.9	22.0	23.5	23.4	16.7	10.3	6.0	3.1		
	2%	WB	0.3	0.4	1.2	4.7	10.8	15.3	18.7	17.8	13.3	8.7	4.3	1.6		
		MCDB	0.5	0.7	2.4	7.6	14.5	19.1	22.2	21.2	15.0	9.3	4.7	2.1		
	5%	WB	-0.8	-0.7	0.3	3.5	9.4	13.8	17.5	16.4	12.4	7.6	2.9	0.7		
		MCDB	-0.6	-0.3	1.2	5.6	13.0	17.3	20.7	19.0	13.9	8.2	3.3	1.1		
	10%	WB	-1.9	-1.9	-0.5	2.5	8.0	12.7	16.5	15.2	11.4	6.6	1.8	0.0		
		MCDB	-1.6	-1.4	0.4	4.2	11.0	15.9	19.4	17.3	12.7	7.3	2.1	0.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.7	6.5	6.0	6.1	6.4	6.2	5.5	4.8	3.5	4.0	4.8		
		MCDBR	3.8	4.2	5.3	7.1	9.3	8.8	8.6	7.2	6.0	3.8	3.1	3.4		
	5% WB	MCWBR	3.8	4.0	4.3	4.5	5.2	4.4	4.0	3.6	3.7	3.1	3.0	3.3		
		MCDWR	4.1	4.2	4.8	6.4	8.6	7.8	7.4	6.5	5.1	3.4	3.0	3.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.214	0.272	0.302	0.333	0.348	0.347	0.371	0.348	0.324	0.289	0.354	N/A		
		taud	1.938	2.067	2.129	2.215	2.389	2.460	2.445	2.484	2.522	2.311	2.615	N/A		
	Ebn at Noon	Ebn at Noon	169	548	738	819	851	862	824	808	745	549	191	0		
		Edh at Noon	15	55	87	105	99	95	94	81	62	43	16	0		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.06	0.52	1.95	3.64	4.67	5.42	4.93	3.42	1.87	0.62	0.12	0.00		
	RadStd	RadStd	0.01	0.09	0.19	0.25	0.34	0.51	0.59	0.32	0.19	0.09	0.01	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	-126	-178	N/A	N/A

Nomenclature: See separate page